PICKFLIP

Design & Implementation of an Intelligent Client-Side Movie Recommendation Engine

Ashwani Kumar (2300520100020) – Lead Aman Chauhan (2300520100012) Vishal Kumar Suman (2300520100077)

Department of Computer Science & Engineering

Supervised by: **Prof. Abhishek Nagar**

June 2026

Executive Summary

The Objective

To eliminate **Choice Paralysis** through a standalone, zero-latency recommendation architecture that operates entirely within the browser runtime environment.

The Innovation

Replacing traditional server-side database clusters with a high-performance, in-memory client repository, synchronized via browser local storage API.

System Architecture

- **ES6 Class OOP Implementation:** Modular structure built using native `MediaItem` constructors and data handling entities.
- **Local Cache Bridge:** Seamless synchronization using browser `LocalStorage` APIs, preserving state without backend hosting bills.
- **Edge Computing Execution:** Executes matching queries in memory in sub-milliseconds, bypassing API latency bottlenecks.

User Discovery Portal (index.html)

Parametric Discovery Mapping

- Clean, responsive dark-themed viewport layout.
- Standardized Genre selection matrices.
- User current mood validation fields.
- Interactive feedback rating buttons.

PickFlip UI Viewport

[Action — Sci-Fi — Comedy]

[Adrenaline — Chill — Mind-Bending]

[Get Recommendations]

Dynamic Score Recalculation Algorithm

To prevent new ratings from instantly skewing historical data points, the engine recalculates a running weighted average:

$$R_{\text{new}} = \frac{(R_{\text{curr}} \times V_{\text{old}}) + R_{\text{user}}}{V_{\text{old}} + 1}$$

- R_{new}: Newly calculated dynamic score display parameter.
- V_{old}: Total historical reviews counter.
- R_{user}: Real-time interactive user score input.

Administrative Portal Gate (admin.html)

Security Checkpoint
Username: [admin]
Password: [********]

[Verify Token]

Token Access Protection

Access to database modification tools is protected via a string-matching gateway. Ensures only authenticated group members can alter internal movie parameters.

Performance Benchmarks

Architecture Setup	Data Location	Average Lookup Latency
Traditional SQL DB	Remote Server Instance	850.0 ms
API Webhook Gateway	Hybrid Network Cluster	320.0 ms
PickFlip Engine	**Client-Side Memory (Edge)**	**0.8 ms**

Running data routines strictly inside local system memory eliminates 99% of network request gaps.

System Verification Matrix

Test Vector	Process Target	Observed Outcome	Status
Genre: Sci-Fi, Mood: Mind-Bending	Filter Loop	Renders "Interstellar" Card	PASSED
Admin Token: "wrong_token"	Authentication	Blocks content, displays error	PASSED
Rating Input: 9.2 Submit	Recalculation Node	Updates local cache structure	PASSED

Project Roadmap & Extensions

Phase 1 (Completed)

Local state OOP logic, browser LocalStorage caching pipelines, and secure Admin Gate portals.

Phase 2

Asynchronous TMDB API integration fetching dynamically millions of global media assets.

Phase 3

Machine learning mood prediction loops and collaborative filtering engines.

Acknowledgements

We express our deep respect and sincere gratitude to our supervisor, **Prof. Abhishek Nagar**, for his technical oversight, structural mentorship, and constant encouragement throughout our development cycle.

Department of Computer Science & Engineering

QUESTIONS?

Thank you for your time.

GitHub Repository:	github.com/Ashwani-3Tech
Live Deployment Platform:	ashwani-3tech.github.io/recommendation-system/
Lead Developer Email:	dharmendrarkt77@gmail.com
Contact Phone:	+91 7084823644